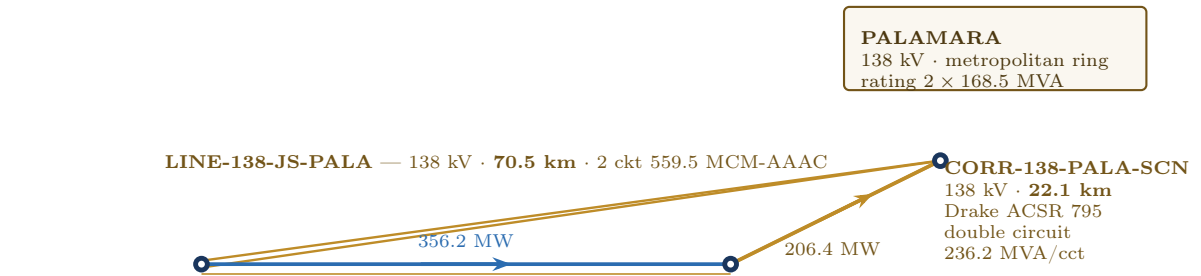
Metropolitan $N-1$ Redundancy Triangle

JSAURI (345/138 kV) — PALAMARA (138 kV) — SAN CRISTÓBAL NORTE (SCN, 230/138 kV) · to-scale one-line geometry



Triangular inequality ($N-1$ redundancy margin)

$d_{\text{JS–PALA}} = 70.5 \text{ km} \;<\; d_{\text{JS–SCN}} + d_{\text{PALA–SCN}} = 50.0 + 22.1 = 72.1 \text{ km}$ (margin: 1.6 km, 2.2 %)

Tripped branch	Pre	ΔP_g / ΔP_{slack}	$V_{\min}$
CORR-138-PALA-SCN	206.4	158.3 / 5.2 MW	SCN
LINE-230-JS-SCN	356.2	141.7 / 3.8 MW	SCN
LINE-138-JS-PALA	—	outside the sealed 8-event artifact	

Pre = pre-event flow [MW]; Δ in MW
Dimensioning event (LINE-345-PC-JS trunk):
687.40 + 58.50 + 6.10 = 752.00